

Intermediate-token windows can restore entity readout

Confirmatory analysis of fixed intermediate-access conditions in Qwen3-8B-Base

Main conclusion

On 36 unseen one-token entities per template, the early intermediate-access window $W_{\text{early}} = [5, 7)$ rescued 25 of 26 *friend* failures with no harm. The sparse intermediate-access set W_{union} rescued all 26 *friend* failures, 8 of 9 *person_in_list* failures, and 18 of 25 *dialogue* failures. Behavioral recovery coincided with restored late final-token entity readout, while the late control window restored neither behavior nor readout.

What this report separates

The full run contains 100 entities, five templates, eight conditions, and 4,000 successful rows. Fourteen entities overlap the pilot. The independent population therefore contains 36 unseen one-token names and 50 unseen multi-token names. These groups answer different questions and are analyzed separately throughout this report.

Population	Role	Primary observation
36 unseen one-token entities	Confirmation of the original window experiment	Prompt-stage bottleneck and window rescue
50 unseen multi-token entities	Separate decoding analysis	First token remains correct, but cached rereading is often needed to complete the name

Run integrity

All 4,000 expected rows completed with zero failures. Hook equivalence, layer-touch policies, cached-generation intervention, clean-reference matching, resume behavior, source hashes, and independent output validation all passed.

Experiment: 2026-09-24 | Model: Qwen3-8B-Base | Greedy generation | 12-token cap | Seed 1729

1. Terms and experimental design

Window notation is zero-indexed and half-open: [5, 7) contains layers 5 and 6. The common name is **intermediate-access window**. The umbrella term for all fixed manipulations is **prespecified intermediate-access conditions**.

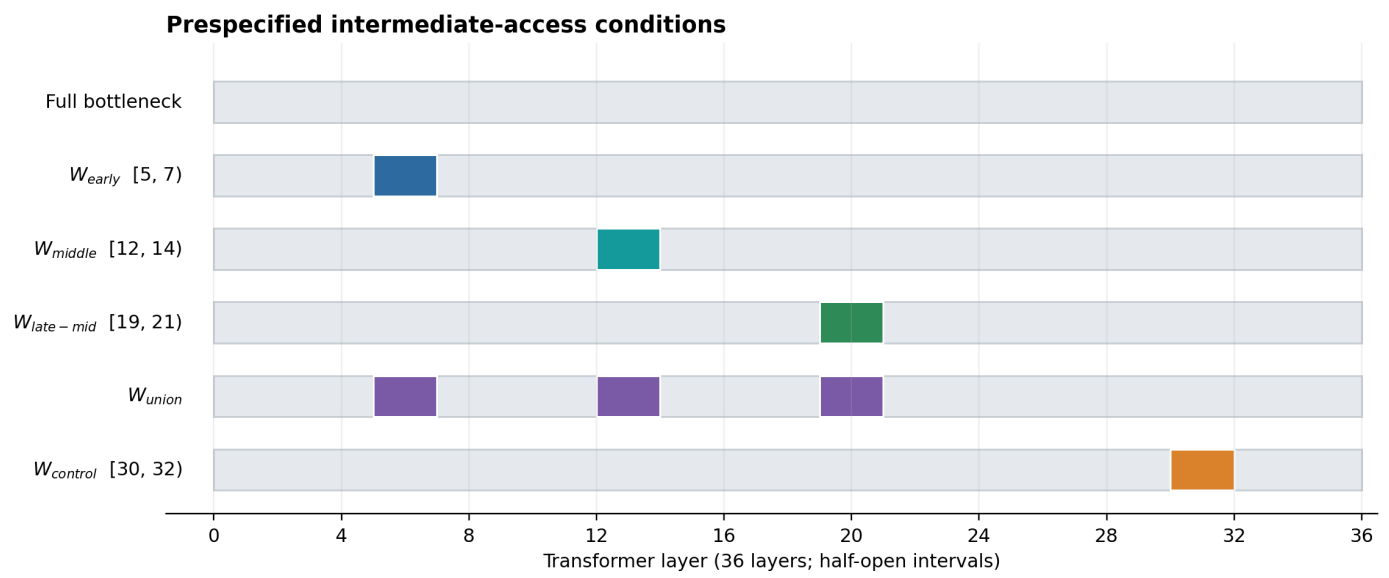
Term	Definition
Intermediate prompt tokens	Tokens strictly after the entity span and before the final prompt token.
Entity-value contribution removal	Keep native QK scores and softmax weights, replace the entity value by zero for targeted queries, and do not renormalize.
Native intermediate access	Intermediate queries receive the original entity-value contribution in the named layer window.
Semantic accuracy	The generated continuation contains the complete target identity. This is the primary outcome.
Exact accuracy	The normalized continuation matches the expected answer exactly. This is secondary because formatting can change.
Rescued	Incorrect under the full bottleneck and correct under a candidate condition.
Harmed	Correct under the full bottleneck and incorrect under a candidate condition.

Prespecified conditions

Symbol / condition	Native intermediate layers	Purpose
Clean	All	Unmodified reference
Generated read block	All prompt layers	Blocks entity rereading only during cached generation
Full bottleneck	None	Removes intermediate entity-value access in all 36 layers
W early	[5, 7) = layers 5-6	Prespecified early candidate
W middle	[12, 14) = layers 12-13	Prespecified middle candidate
W late-mid	[19, 21) = layers 19-20	Prespecified later-middle candidate
W union	{5, 6, 12, 13, 19, 20}	Sparse set combining the three candidates
W control	[30, 32) = layers 30-31	Matched late negative control

Experimental procedure

Each entity-template pair was evaluated under the same eight fixed conditions. The intervention changes only the entity-value contribution available to intermediate prompt-token queries; native attention scores and all other value contributions are preserved.

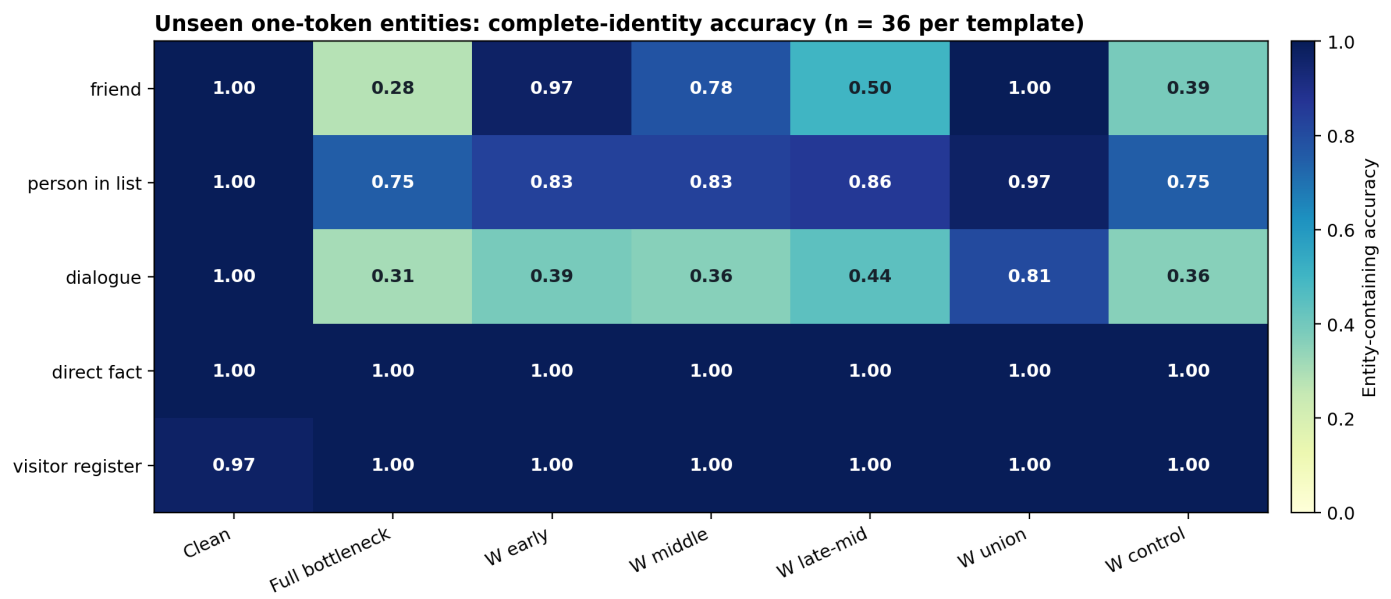


Step	Operation
1. Build the prompt	Insert one of 100 entity names into each of five templates and identify the entity span, intermediate prompt tokens, and final prompt token.
2. Apply a condition	For the full bottleneck, remove entity-value contributions to every intermediate query. For a window condition, restore native access only in its named layers.
3. Generate	Keep the final prompt query native. During cached generation, block new-token queries from rereading the entity in every non-clean condition.
4. Score and pair	Measure complete-identity semantic accuracy, then compare each candidate with the full bottleneck on the same entity-template pairs.

Gray indicates layers where intermediate entity-value access is removed; colored spans indicate restored native access. The late control window has the same width as each candidate but lies near the end of the model.

2. Confirmation on unseen one-token entities

This is the population that matches the original pilot's one-token setup. Generated-token blocking alone matches clean semantic accuracy across all five templates, reproducing the earlier observation that the answer identity is established at the final prompt state for one-token names.



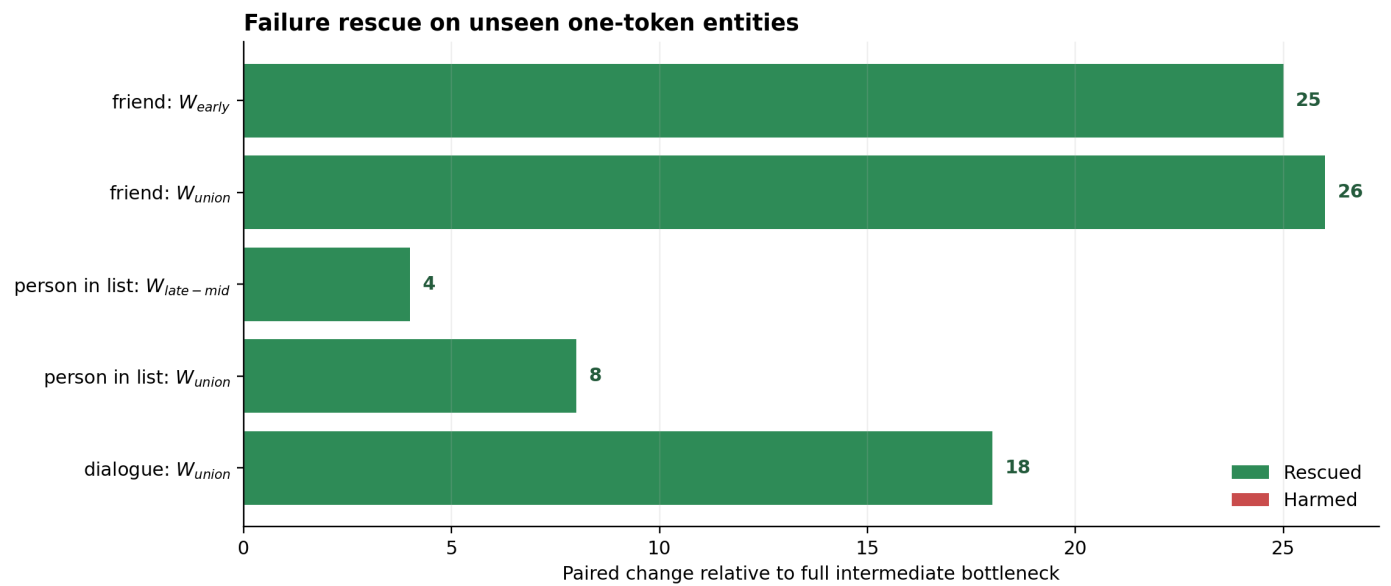
Template	Clean	Full bottleneck	Template-primary	W union	W control
friend	1.000	0.278	0.972 (W early)	1.000	0.389
person_in_list	1.000	0.750	0.861 (W late-mid)	0.972	0.750
dialogue	1.000	0.306	0.806 (W union)	0.806	0.361
direct_fact	1.000	1.000	1.000 (W middle)	1.000	1.000
visitor_register	0.972	1.000	Insensitive control	1.000	1.000

Interpretation

The full intermediate bottleneck strongly damages *friend* and *dialogue*, and moderately damages *person_in_list*. The prespecified early window nearly restores *friend*; the sparse union restores all *friend* failures and most failures in the other two sensitive templates. *direct_fact* and *visitor_register* are semantic controls because the bottleneck does not reduce complete-identity accuracy.

3. Failure-level rescue

Paired analysis asks whether each entity that failed under the full bottleneck becomes correct under a fixed candidate condition. This avoids understating effects in templates that begin near the ceiling.



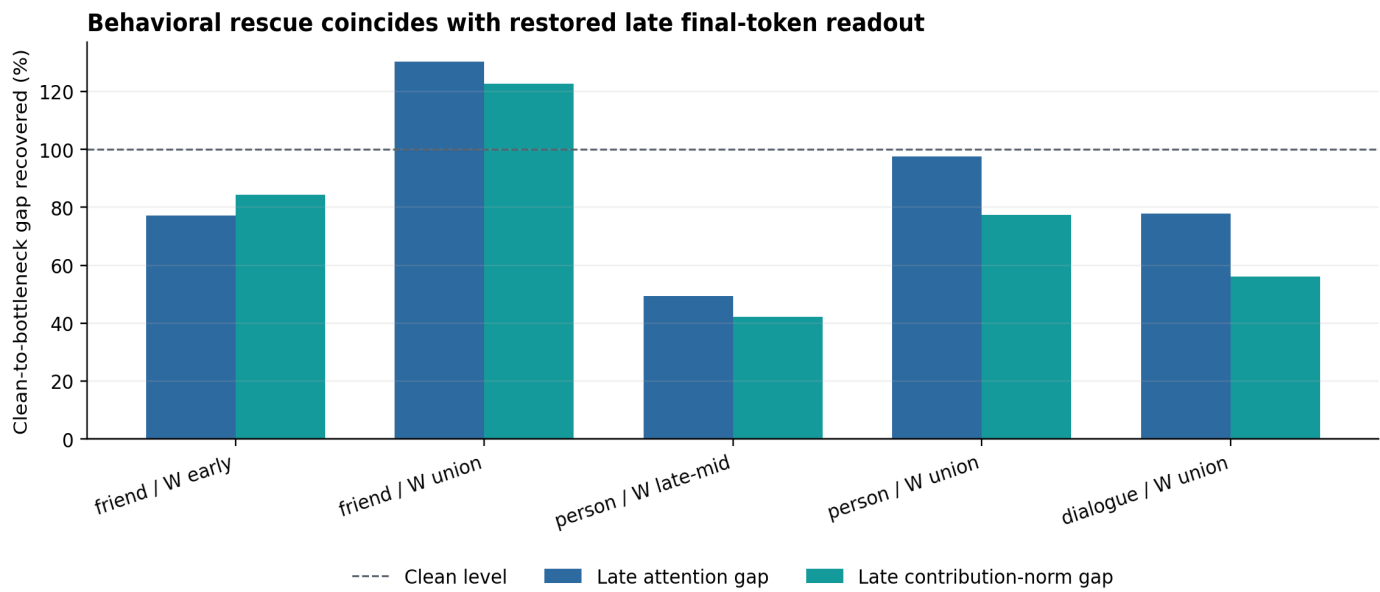
Comparison	Full	Candidate	Rescued	Harmed	Exact paired p
friend: W_{early}	0.278	0.972	25 / 26 failures	0	5.96e-08
friend: W_{union}	0.278	1.000	26 / 26 failures	0	2.98e-08
person_in_list: $W_{late-mid}$	0.750	0.861	4 / 9 failures	0	0.125
person_in_list: W_{union}	0.750	0.972	8 / 9 failures	0	0.0078
dialogue: W_{union}	0.306	0.806	18 / 25 failures	0	7.63e-06

The strongest single-window confirmation is W_{early} for *friend*: 96% of bottleneck failures are rescued without harming any bottleneck-correct entity. $W_{late-mid}$ for *person_in_list* is directionally positive but individually underpowered. W_{union} provides significant recovery for *friend*, *person_in_list*, and the previously unseen *dialogue* template.

The late control remains near the full bottleneck. This makes generic activation from opening any two layers an unlikely explanation.

4. Mechanistic evidence: restoration of late readout

The intervention never modifies the final prompt query directly. Nevertheless, successful early and middle windows restore the final token's late attention to the entity and its projected entity-contribution norm in layers 24-35. Because causal masking prevents later intermediate tokens from changing the earlier entity representation, this pattern is consistent with intermediate tokens configuring the final token's later query and readout state.



Template / condition	Attention gap recovered	Contribution-norm gap recovered	Semantic accuracy
friend / W early	77%	84%	0.972
friend / W union	130%	123%	1.000
person_in_list / W late-mid	49%	42%	0.861
person_in_list / W union	98%	77%	0.972
dialogue / W union	78%	56%	0.806
Late control	approximately 0%	approximately 0%	Near full bottleneck

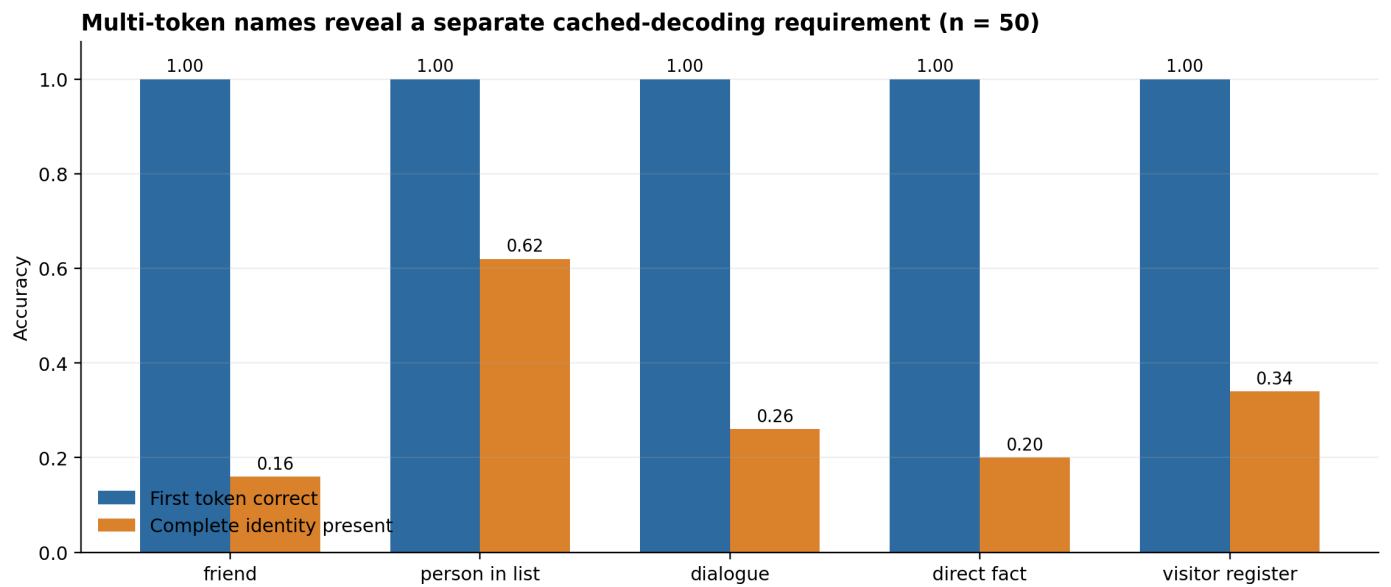
Supported causal account

Entity value -> intermediate-token computation in selected early/middle layers -> final-token state/query configuration -> stronger late direct entity readout -> correct identity generation.

The data support sufficiency of small, prespecified routes. They do not show that any one interval is uniquely necessary or optimal.

5. Multi-token names answer a different question

Half of the 100 entities contain two or more tokens. The runner accepted them by treating the first answer token as the next-token target while retaining complete-name scoring. Under generated-token blocking, the first token remains correct, but subsequent tokens often fail because cached generation cannot reread the entity span.



Template	First token correct	Complete identity present	Most common failure
direct_fact	1.00	0.20	First name only (40/50)
person_in_list	1.00	0.62	First name only (19/50)
friend	1.00	0.16	First name only (41/50)
visitor_register	1.00	0.34	First name only (32/50)
dialogue	1.00	0.26	First name / malformed continuation

Examples

Target identity	Generated-read-block completion
Emily Dickinson	Emily
Mary Shelley	Mary
Oscar Wilde	Oscar

This is an additional result: the final prompt state can establish the first name token, while correct multi-token completion often requires entity rereading during autoregressive decoding. These cases should not be pooled with the one-token confirmation when estimating window rescue.

6. Conclusions and manuscript use

Primary finding

Small, prespecified intermediate-access windows are sufficient to restore entity extraction in sensitive one-token prompts. The strongest confirmation is the early window $W_{\text{early}} = [5, 7)$ for *friend*. A sparse union of six early-to-middle layers generalizes recovery to *friend*, *person_in_list*, and *dialogue*.

Mechanistic implication

Recovery of late final-token attention and projected contribution norm indicates that intermediate tokens help configure a later direct readout from the original entity. The near-null late control separates this effect from merely reopening an arbitrary pair of layers.

Recommended manuscript statement

On unseen one-token entities, native intermediate access in layers [5, 7) rescued 25 of 26 friend failures without harming any correct case. A prespecified sparse access set spanning layers [5, 7), [12, 14), and [19, 21) rescued all 26 friend failures, 8 of 9 person_in_list failures, and 18 of 25 dialogue failures. Behavioral recovery tracked restoration of the final token's late entity-directed attention and value contribution, whereas a matched late control showed essentially no readout recovery.

Boundaries

Supported	Not established
Sufficiency of fixed early/middle intermediate-access routes	Unique necessity or exact optimal boundaries
Generalization across three sensitive prompt forms	Universal recovery across every template
Late readout restoration accompanies behavioral rescue	A single exclusive token-to-token relay path
Cached rereading supports multi-token name completion	Prompt-stage window effects for multi-token names without a dedicated rerun

Recommended next experiment

For multi-token names, rerun the prompt-stage intermediate conditions while keeping generated-token entity reads native. This separates prompt-stage configuration from the separate requirement to complete later name tokens during cached generation. If the paper requires a 100-entity replication of the original task, use 100 genuinely one-token entities.

Provenance

Source run: 12-intermediate-window-confirmation-entities100_20260924_012450. Raw results: results.jsonl. Validation: validation.json and independent_validation.json. Pilot overlap: pilot_overlap.json. Runner: qwen_intermediate_window_confirmation.py.